Teo Milos

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teomilos.ca

EDUCATION

DALHOUSIE UNIVERSITY 2019 - 2023

Bachelor of Engineering (B. Eng) in Electrical Engineering

- Expected graduation date: December 2023

PROFESSIONAL EXPERIENCE

AETHERA TECHNOLOGIES May – August 2023

ELECTRICAL ENGINEERING INTERN

- Set up SNMP to Modbus gateway registers for global data collection across multiple transmit/receive sites, employing digital logic and utilizing MIB Browser
- Developed test protocols and conducted tests for the Human-Machine Interface (HMI) of solid-state RF heating generators, as well as modulators and demodulators

NOKIA September – August 2022

HARDWARE ENGINEERING INTERN

- Performed high accuracy voltage ripple measurements on IP routers and assessed the compliance of power supply noise using Cadence Allegro PCB designer, benchtop electronic test equipment, Matlab, and SMD soldering
- Wrote a Perl script to control the voltage and current characteristics of several power supplies

GEOSPECTRUM TECHNOLOGIES INC January – April 2022

ELECTRICAL ENGINEERING INTERN

- Designed circuits to attenuate the drive signal on a sound projector based on the input current using comparator registers, assembly code, and integrated circuits
- Implemented digital equalization filters to compensate for sound projector frequency response
- Made circuits to mimic the voltage output of a pressure sensor at multiple depths

SIEBEN LABORATORY - DALHOUSIE UNIVERSITY May – August 2021

RESEARCH ASSISTANT

- Re-designed PCBs using Eagle and KiCad to be sued as integral parts of microfluidic phosphate sensors
- Contributed to the development of a lab-on-chip fish welfare sensor for application in aquaculture using enzyme-linked assay (ELISA)
- Wrote research-related technical documents (LaTeX)

RELEVANT PROJECTS

ROBOTIC ARM May 2023 - Present

- Building a 3D printed 6-DOF robotic arm, incorporating Arduino and GRBL for precise control
- Developing kinematic algorithms, integrating sensors, and fine-tuning movements to enable human-like actions

AUTONOMOUS ROBOT May – July 2022

- Built an autonomous robot capable of obstacle avoidance, disposal, and detection of simulated mines using Robot Operating System (ROS), Linux command line, shell scripting, and version control (GIT)
- Second place

UNDERWATER REMOTELY OPERATED VEHICLE (ROV) September 2016 – June 2019

- Designed an electronic system, control scheme, and utilized 3D printing, Python, C/C++, and Arduino to create an ROV that meets the needs of the Eastman Company
- Proposed a business pitch, managed budget, and gained team sponsors
- Advanced to Marine Advanced Technology Education (MATE) International ROV Competition

NOVEL INTELLIGENT IRRIGATION SYSTEM January – May 2018

- Designed and programmed an Intelligent Irrigation System via Python, Raspberry Pi, and Matlab, capable of saving resources and responding based on various sensor inputs
- Bronze medalist at Canada-Wide Science Fair

MICHELSON'S INTERFEROMETER January – May 2019

- Utilized the principle of interference to construct an interferometer and determine the refractive index of glass

VOLUNTEERING & EXTRA CURRICULARS

- **PRESIDENT** (May 2021 – Present): Dalhousie Women in Engineering Society
- **MAKER** (September 2023 – Present): Makers Making Change
- **MEMBER** (May – December 2022): Dalhousie Rocketry Team
- **MEMBER** (May – December 2021): Dalhousie CubeSAT Satellite Team
- **MENTOR** (February – April 2021): COVE's Young Women in Ocean Tech Program
- **CAPTAIN & PROGRAMMER** (2016 – 2019): South Shore Robotics

AWARDS & HONORS

- **SECOND PLACE** (2022): Dalhousie Peter Gregson Robotics Competition
- **NSERC UNDERGRADUATE RESEARCH AWARD** (2021): The most prestigious research award in Canada
- **DEAN'S LIST, SEXTON SCHOLAR** (2020 - 2021): GPA > 3.85, top 8-10% of engineering students
- **MOST INNOVATIVE** (2019): South Shore Robotics
- **BRONZE MEDALIST** (2018): Canada-Wide Science Fair